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Analysis and visualization of COVID-19 discourse on Twitter using data science: a case study of the USA, the UK and India

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Abstract

Purpose – This paper aims to understand, examine and interpret the main concerns and emotions of the people regarding COVID-19 pandemic in the UK, the USA and India using Data Science measures.

Design/methodology/approach – This study implements unsupervised and supervised machine learning methods, i.e. topic modeling and sentiment analysis on Twitter data for extracting the topics of discussion and calculating public sentiment.

Findings – Governments and policymakers remained the focus of public discussion on Twitter during the first three months of the pandemic. Overall, public sentiment toward the pandemic remained neutral except for the USA.

Originality/value – This paper proposes a Data Science-based approach to better understand the public topics of concern during the COVID-19 pandemic.

Keywords Topic modeling, Sentiment analysis, COVID-19, Twitter, Data science, Machine learning

Paper type Research paper

1. Introduction

The COVID-19 contagion has swept across the world claiming lives of around a million individuals to date. Declared a pandemic on March 11, 2020 by the World Health Organization, this virus has brought countries to a standstill and has disrupted human life, the likes of which we have not seen in decades. The impact and disruption brought about by this virus has captured headlines making it one of the most talked about topics of discussion. Not only has it brought a drastic loss of life, but it has also repressed societies and upended the world economy. This has led to a great deal of research on COVID-19.

One domain witnessing large amount of COVID-19–related research is that of social media analysis (Depoux *et al.*, 2020; Gao *et al.*, 2020; Pennycook *et al.*, 2020). Twitter has enabled researchers' access to real-time user data, allowing them to gather opinions of millions of users (Bozanta and Kutlu, 2018; Yaqub *et al.*, 2020). Analysis over these data sets, containing rich user information, helps them understand public opinion regarding the

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pandemic (Elkin *et al.*, 2017). The sentiment and topics of tweets may be mined for indications of citizen opinion and supplementary information regarding COVID-19 events and conditions.

Although many studies look at tweets for sentiment and behavior regarding the medical progression of the pandemic (Mackey *et al.*, 2020; Rosenberg *et al.*, 2020), this study examines citizen sentiment toward the topics of their concern. We collected around 1 million tweets from three large diverse countries, namely, the USA, the UK and India for three months starting from March 1 to May 31, 2020. We first identify topics in our data set using topic modeling, an unsupervised machine learning technique. We then evaluate sentiment of some of these topics, which occur frequently among discussions in all three countries included in our study. Obtaining such an understanding of user reactions is important to governments, policymakers, media, researchers and fellow citizens. Hence, the purpose of this study is to investigate the following questions:

Q1. What were the main topics of concerns for Twitter users toward COVID-19?

Q2. What is the public sentiment toward these prominent topics?

The rest of the paper is structured as follows: In Section 2, we review existing literature, whereas in Section 3, we formulate our research questions. In Section 4, we discuss our methodology, whereas in Section 5, we describe our findings. In Section 6, we discuss our analysis results, whereas finally, in Section 7, we conclude our paper.

2. Literature review

With the increasing digitalization of society, there has been a dramatic increase in the amount of Twitter activity in recent years. This has resulted in the availability and collection of huge amounts of Twitter data for research studies in various fields ranging from health care to business analytics. In the past, Twitter feeds have been used to analyze the political discourse of public discussion in the context of a presidential election and events of high political significance such as Brexit (Ilyas *et al.*, 2020; Yaqub *et al.*, 2017). In the field of health informatics, investigators have leveraged natural language processing techniques to understand the psychological well-being of users, medicine intake and chemotherapy (Mahata *et al.*, 2018; McClellan *et al.*, 2017; Zhang *et al.*, 2018). In business analytics, there are examples of case studies of Starbucks' brand authenticity and Uber's customer engagement created using text analytics techniques on Twitter data (Khetan *et al.*, 2019; Shirdastian *et al.*, 2019). Twitter data has also been used by researchers to investigate the discourse of public discussion after natural disasters. For example, (Wang and Zhuang, 2017) used Twitter feed to analyze crisis communication during 2016 Hurricane Sandy and (Subba and Bui, 2017) during 2015 Nepal earthquake.

Twitter data has been used by researchers for studying information dissemination in past health crises. For instance, during the Ebola outbreak in 2014, investigators found spatial and social distance to be an important predictor for measuring public attention and fear toward the disease (Van Lent *et al.*, 2017). In other studies, researchers conducted longitudinal analysis of the public blame expressed toward governmental institutions alongside the spread of Ebola (Roy *et al.*, 2016). Some investigators also applied topic modeling and sentiment analysis to compare the information spread on social media and mainstream news outlets (Kim *et al.*, 2016). Similarly, following the outbreak of Middle East respiratory syndrome in South Korea, various studies were found Twitter streams to be a good indicator of not only measuring the outspread of the disease but also devising assistance policies by health authorities (Do *et al.*, 2016; Shin *et al.*, 2016).

Regarding the current COVID-19 pandemic, researchers were quick to react to the ongoing developments and immediately started the process of data archiving, which allowed many researchers to publish studies in a short span of time. Dong *et al.* (2020) at the Johns Hopkins University were the first to create an interactive dashboard to map the spread of the virus. Shortly afterwards, Chen *et al.* (2020) created the first publicly available multilingual Twitter data set related to COVID-19 to study the impact of the virus on online user behavior. Various investigators were able to discover a strong shift in user sentiment from fear to anger owing to shortages of COVID-19 testing and medical supplies in the early stages of the pandemic (Lwin *et al.*, 2020; Samuel *et al.*, 2020).

Although the top-trend feature of Twitter can give us a cursory look at what people are discussing, this simple filter is not sufficient to capture the main public concerns. Therefore, many researchers must bridge this gap by applying unsupervised machine learning techniques, such as topic modeling, to extract the underlying topics being discussed by the public. For example, Sha *et al.* (2020) used dynamic topic modeling on tweets made by several US governors and presidential cabinet members and found a transition in narrative from low risk to high risk regarding the pandemic. Ordun *et al.* (2020) made use of this technique to do exploratory analysis of COVID-19. Similarly, Kabir *et al.* (2020) leveraged this technique to create an application to isolate and visualize different subtopics on Twitter feeds in the context of the USA, whereas Abd-Alrazaq *et al.* (2020) were able to observe different themes of discussions such as postponement of sports events, closure of fast-food chains and impact of the pandemic on the US economy.

Although Twitter is a good tool for social media engagement, it is often exploited for promulgating the spread of misinformation. This problem is exacerbated especially when misinformation is related to public health. A significant number of information science researchers have termed COVID-19 as an "Infodemic" (Cinelli *et al.*, 2020). Hence, much contemporary research has also been conducted to gauge the dissemination of false information and conspiracy theories related to COVID-19. For instance, Kouzy *et al.* (2020) found an alarmingly high rate of spread of misinformation and unverifiable content related to COVID-19 on Twitter. Singh *et al.* (2020) not only found correlation of new cases with Twitter activity but also discovered high prevalence of non-credible sources in Twitter feeds. Similarly, Ferrara (2020) identified bots (automated Twitter accounts) that were using ideological hashtags for spreading conspiracy theories about COVID-19. Likewise, Ahmed *et al.* (2020) created social network graphs for identifying the sources of 5 G COVID-19 conspiracy theory on Twitter in the UK and (Budhwani and Sun, 2020) discovered the spread of "China-virus" chanter by alt-right groups.

3. Research questions formulation

Following the outbreak of COVID-19, different countries enacted different response measures. In addition, there was proliferation of misinformation regarding the pandemic on social media, which often overshadowed the guidelines of health officials and the concerns of the common people. Furthermore, there was constant discord between policymakers and health officials especially in the USA regarding the coping methods for containing the spread of the virus. Amidst all this confusing hyperactive discussion, it became difficult to hear the main concerns of the people. Hence, we propose the following research question:

RQ1. What were the main topics of discussion on Twitter related to COVID-19?

Public sentiment generated by applying natural language processing algorithms on Twitter data is often considered an apt indicator of the public's actual opinion and yields interesting insights. The lockdown measures taken by several governments spurred the usage of social media, especially Twitter, which provided people an ideal platform to express their emotion regarding the different factors revolving around the pandemic. In the absence of physical opinion mining tools, Twitter data is ideal to determine the polarity of people regarding different aspects of the pandemic. However, as there were so many points of discussion around the pandemic and there were new developments happening each day, it was difficult to narrow down to the topics that were most pertinent in portraying the public's emotions. In addition, there was a varied response by many countries which could be attributed to the domestic policies, spread of the virus and health infrastructure. Hence, to cater to this issue, we look at the topics generated by our topic modeling algorithm and select only those that are conspicuous and consistent in all three of the countries. We believe that this analysis of public sentiment can aid in understanding the public perspective during the pandemic. Hence, we propose the following research question:

RQ2. What was the public sentiment toward topics that were consistent in different countries?

4. Research methodology

4.1 Data

For this paper, we collected tweets for a 92-day time period spanning from March 1 to May 31, 2020, using the keyword "coronavirus" in the Twitter streaming application programming interface (API). We retained only those tweets that were in English. Afterwards, we separated the tweets based on their respective location attributes returned by the streaming API into three geographical locations, i.e. the USA, the UK and India. Toward the end, we had over 320,000 tweets for the USA, 340,000 tweets for the UK and over 310,000 tweets for India. For our analysis, we needed only the time stamp, text and location attribute of the tweet. Therefore, we removed all the other metadata from the collected tweets. We believe that this geographic segmentation of tweets is very important as the spread of the pandemic and the government's response was different in many countries. Furthermore, we choose these countries because they accounted for most tweets in our data set. In addition, these three countries being part of different continents help us in doing a holistic analysis of the pandemic around the globe. The tweets data set used in this study is publicly available [1] to any researcher who wishes to replicate our results.

4.2 Topic modeling

Topic modeling is a probability based unsupervised learning technique, which not only extracts the semantic topics but also outlines relative importance of each topic in the text (Blei *et al.*, 2003). For our analysis, we used the Gensim library [2] developed by Rehurek and Sojka (2010) in Python, which makes use of the Latent Dirichlet Allocation algorithm (LDA) – one of the most popular text mining algorithms (Jelodar *et al.*, 2019). For our analysis, we filtered tweets from our data set based on three different locations: the USA, the UK and India. The topic model was then used to extract topics individually for each location.

The LDA algorithm is based on the assumption that in a collection of interrelated documents, there are some combinations of topics included in each document. The first step toward using LDA is to clean the tweet text by removing uniform resource locator, emails addresses and emojis. The next step is to remove stop words such as "a," "her" and "him" from the text. For this purpose, we started off with the standard stop words list given in Python's wordcloud library. Later on, we added some context specific words to that list such as "covid" and "coronavirus."

The next step is to perform text lemmatization, which is an application in natural language processing that converts a particular word to its root meaning, thereby reducing the complexity of the words. We have used Python's Gensim library lemitizer for this purpose. The final step is feeding this bag of lemmatized words to the Gensim model, which then extracts the topics.

The number of topics to extract is a parameter that needs to be set by the user. To determine this parameter, we ran the topic model with varying number of topics and calculated the corresponding coherence scores. Coherence measures score a single topic by measuring the degree of semantic similarity between high scoring words in the topic. Figure 1 shows the coherence scores for different number of topics for each data set. From these results, we observed that the highest coherence scores were observed when five topics were set for both the USA and India with scores of 0.403 and 0.396, respectively. For the UK data set, five topics do not yield highest coherence score; nonetheless, this score of 0.384 is marginally lower than four or six topics. Hence, we decide on five topics to be ideal in optimally representing the COVID-related discussion on Twitter.

4.3 Sentiment analysis

Sentiment analysis is a text mining technique that extracts the underlying emotion in each text. This technique has many applications. However, because of the short length of tweets and inclusion of various slang and emojis, it becomes cumbersome to accurately determine the underlying sentiment using traditional sentiment analysis techniques. Therefore, for this study we used Vader library (https://pypi.org/project/vaderSentiment/) created by Gilbert and Eric (2014) in Python, which is a lexicon and rule-based sentiment analysis tool specifically tailored to extract sentiment from social media text. Vader does not require any training data but is constructed from a generalizable, valence-based, human-curated gold standard sentiment lexicon.

A major advantage of Vader is that it does not give a hard classification of tweets such as positive or negative. Instead, it gives out the proportion of text that falls into each sentiment category. Moreover, it gives out a compound score that tells us the overall sentiment polarity of the text. The compound score ranges between −1 and +1, which +1 being the most

Figure 1. Topic modeling coherence scores

Note: We can observe five topics as having highest coherence scores for the USA and India positive and −1 being the most negative. In our analysis, we have mainly focused on the compound score of tweets. Furthermore, another advantage of this technique is that it does not require extensive text preprocessing.

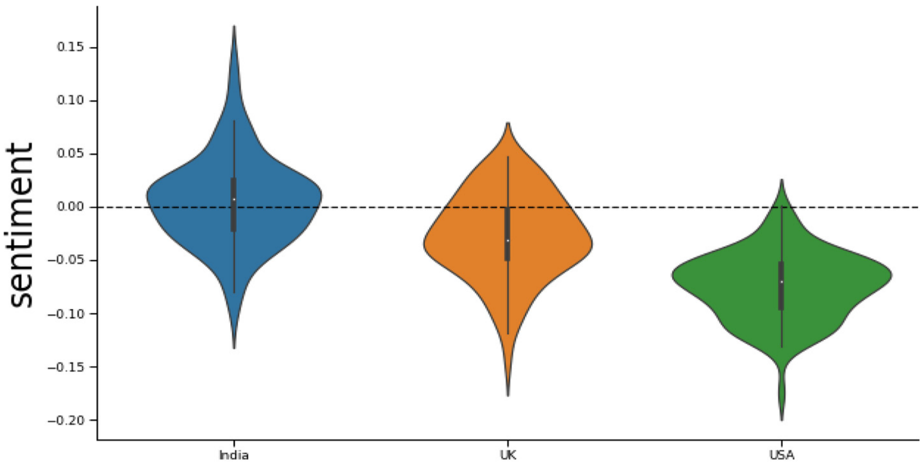
5. Findings

5.1 Initial exploratory data analysis

From the word clouds shown in Figure 2, we can see that the discussions in all three countries predominantly contain various COVID-19–related terms such as pandemic, cases, deaths, testing and hospital. The mention of China is also prevalent among the conversations. Furthermore, we can also observe that the conversations surrounding COVID-19 have a political aspect to them. Words such as government and state can be seen and political leaders such as Trump, Boris Johnson and Modi are all frequently mentioned.

The violin plot in Figure 3 shows the distribution of the daily mean sentiment value for each of the three countries. The width of each violin represents the probability density of the sentiment across all its values. We can also see a boxplot within the violin that indicates

Figure 2. Word clouds of Twitter messages from India, the UK and the USA



Notes: Breath of the plot indicates number of tweets with respect to sentiment on the y-axis. Here, majority of tweets from India had a positive sentiment, whereas on average, tweets from the UK and the USA had negative sentiment

Figure 3. Violin plot for sentiment scores for India, the UK and the USA

summary statistics such as the median and interquartile range. As indicated by the violin plot, the sentiment distribution pertaining to COVID-19 varies across all three countries. Discussions in the USA are far more negative as compared to India and the UK. This is illustrated by the fact that the mean sentiment values for the USA remains negative on all days throughout the 92-day period. Similarly, although some tweets generated from the UK cross over to the positive side, the overall sentiment remains predominantly negative. In contrast, the violin plot for India remains balanced with a seemingly equal proportion of positive and negative tweets.

5.2 RQ1: topic modeling of tweets

We extracted the five topics for each country and analyzed them by observing the word cloud, frequency and sentiment of each extracted topic.

5.2.1 United Kingdom. The word cloud in Figure 4 shows the main keywords for each topic. We can see how words that tend to be used together in real-life conversations are grouped together in the same topic by our model. For instance, topic 3 contains words such

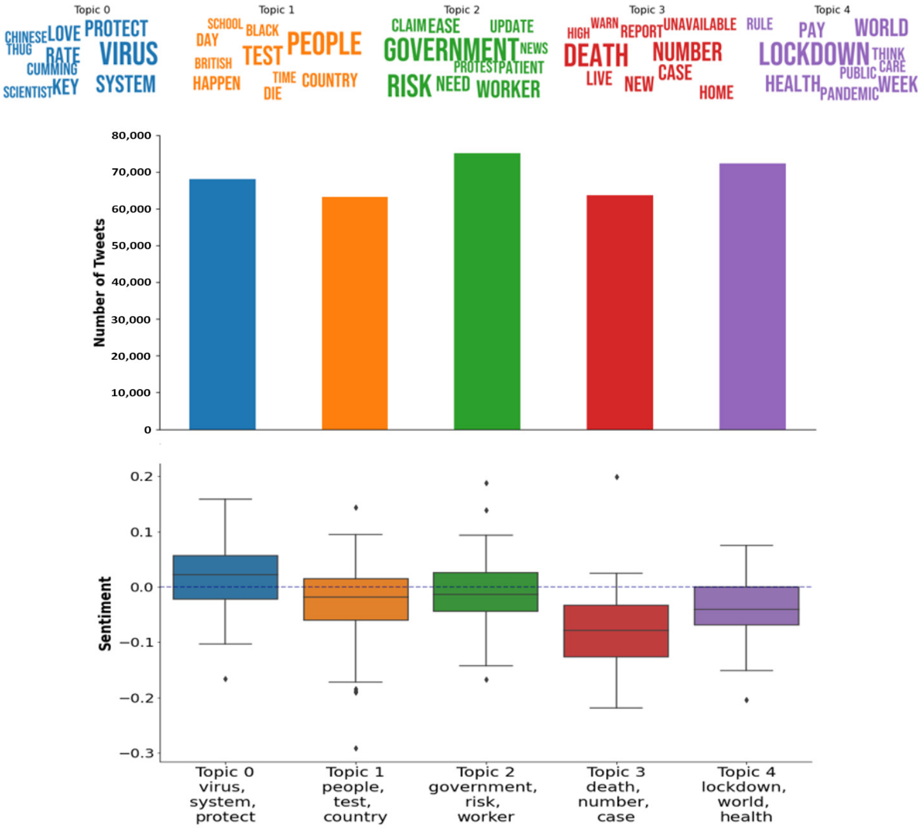
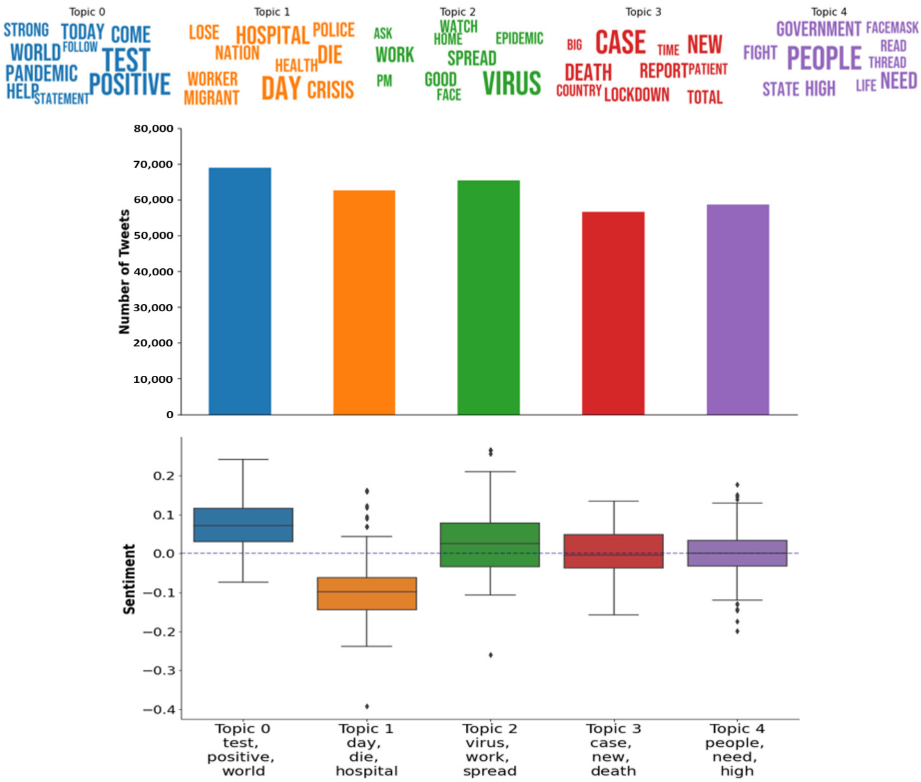


Figure 4.
Topic modeling results for the UK

Notes: Highest number of tweets relate with topic 2, as shown in the bar graph. The violin plot indicates topics 0 and 3 with the most positive and negative sentiment, respectively as new, death, case, report, number and high. As these words are commonly used in conversations discussing the spread of the virus, we can label this topic as the spread of virus. Furthermore, the bar chart shows the number of tweets for each topic. We can see that topic 2, which discusses the government, is the most popular topic in tweets generated from the UK. From the boxplot, we can see the daily mean sentiment distribution for the tweets in each topic. The sentiment varies from topic to topic. It is clear that topic 3, discussing the virus spread, is the most negative topic having a negative sentiment on nearly all the days of our data set. Topic 2, discussing the government, is also negative on a majority of days. However, it does have a positive sentiment on some days.

5.2.2 India. Figure 5 shows us the main keywords for each topic in tweets from India. Similar to the UK, we can see that topic 3 discusses the spread of the virus. It contains terms such as new, death, case, report and total. Second, we observe that topic 0 is the most popular topic in India, followed by topics 2 and 1. In contrast to the UK, we can see that the topic discussing the government is not as popular. Finally, we can see that the sentiment values are either positive or neutral on a majority of days for all topics except topic 1. Topic 1, which discusses words such as hospital, police, health and crisis, is by far the most negative topic in India.



Notes: Highest number of tweets relate with topic 0. The violin plot indicates topics 0 and 1 with the most positive and negative sentiment, respectively

Figure 5.
Topic modeling results for India

5.2.3 United States of America. From Figure 6, we discover the main keywords used in each topic for discussions in the USA. Topic 1 discusses the spread of the virus and contains terms such as new, case, death, report and die. In addition, we can see that the most popular topic is topic 0, which discusses people and probably the origin of the virus. In comparison, the topic that discusses the government is much less frequent. On the other hand, the topic that contains Trump was also discussed very frequently. Furthermore, we can observe that the sentiment of all topics predominantly remains negative. This is different from India and the UK, where sentiment of some topics crossed over to the positive side, indicating that people in the USA exhibited more negative response toward the pandemic.

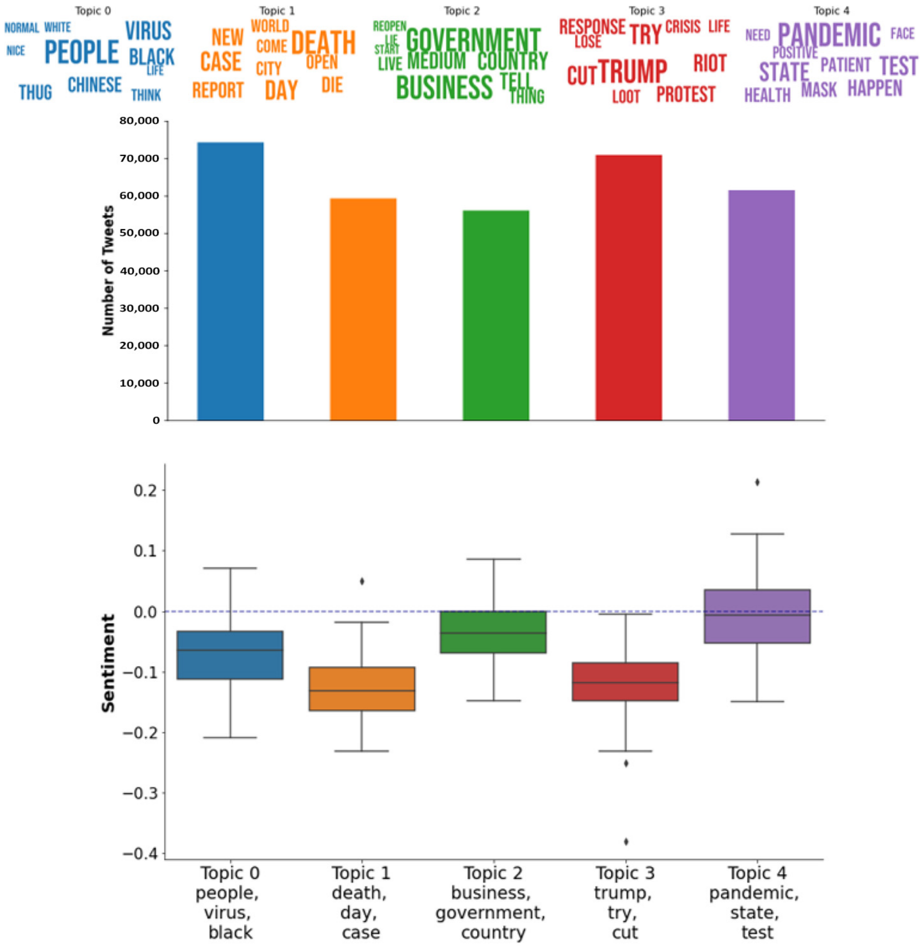


Figure 6.
Topic modeling results for the USA

Notes: Highest number of tweets relate with topic 0. The violin plot indicates topics 4 and 1 with the most positive and negative sentiment, respectively

5.3 Rq2: public sentiment toward the topics that were consistent in different countries

In this section, we analyze the daily mean sentiment values over time for the topics pertinent to the government and the spread of pandemic in each country by using the topics generated from the previous research question. Figure 7, below, shows that although sentiment values toward government topic in India and the UK fluctuate between positive and negative; these remain negative across all 92 days in the USA. From Figure 8, which demonstrates the public sentiment toward the topic discussing the spread of the virus, we also observe a similar pattern for each of the three countries.

We then compared the peaks and drops in sentiment to some of the decisions taken by the government at the time and found some interesting results. On 20 March, for instance, the UK Government's Coronavirus Job Retention Scheme was followed by a sharp rise in government-related sentiment in the UK. Moreover, on 29 May, when the end of the scheme was announced, we observe a sudden drop in sentiment. For the USA as well, we see a drop in sentiment toward the end of March, which coincides with the time when California issued a Stay-at-Home Order. We also see a sudden rise in sentiment in late May, which is when the US Government and AstraZeneca announced a collaboration to aid the development of a possible COVID-19 vaccine. Similarly, in India, we see a rise in sentiment in late March, which can be attributed to the Indian Prime Minister's request to all citizens to applaud as a symbolic gesture to show support for people working in essential sectors in times of the virus.

6. Discussion

This research paper discusses two research questions. The first research question uses topic modeling to investigate the main subjects of discussion on Twitter related to COVID-19. We have extracted two values for each country, the first shows the number of tweets by dominant topics and the second value shows the overall sentiment for each topic. In the UK, we find that the government topic is dominating the other topics with more than 7,000 tweets. However, its sentiment tends to be neutral, which meant that people in the UK were

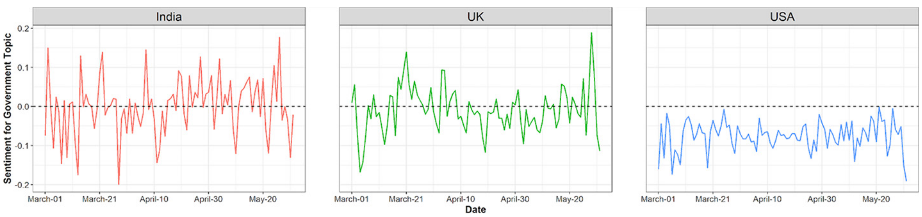


Figure 7. Average daily public sentiment in the three countries toward "Government" topic

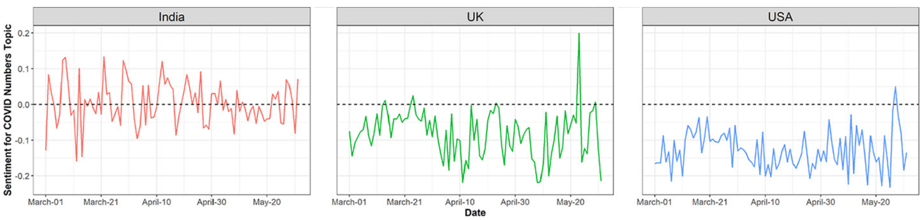


Figure 8. Average daily public sentiment in the three countries toward "Spread of COVID-19" topic

less critical of their government's pandemic strategy. On the other hand, topic 3, which is about virus spread, provides the most negative sentiment, which depicts people's concerns regarding the spread of the virus. In India, we find that topic 0, which is about testing for the virus, is the most popular topic followed by topic 2, which is talking about virus spread. The government topic is not as prevalent in India. Furthermore, topic 1, which is about death and hospitals, has by far the most negative sentiment, which can be attributed to the rising virus spread. In the USA, we can notice that the most popular topic is topic 0, which contains tweets about different conspiracy theories regarding the origin of the virus, which depicts the polarizing nature of the discussion on the pandemic. Interestingly, the government topic is the least frequent among other topics, which exhibits that public discussion on pandemic, did not focus entirely on the government. Additionally, the sentiment of all five topics in the USA remains negative, which is different from India and the UK where some topics display positive sentiment.

The second research question investigates public sentiment toward the two most frequent topics of discussion identified through topic modeling. The two topics used for this analysis are government and the spread of COVID-19. Although we suspect that other terms such as Trump (government leaders) could serve as proxies for the government term, we have not acquired an extensive list of other potential proxies and therefore hesitate to add these few other additional terms. We analyze the daily mean sentiment toward government for the three countries. We notice that in the USA, the sentiment stays negative all the time during the 92 days. However, the sentiment values for the UK and India fluctuate between negative and positive. We also find that the sentiment for each country using the government-related tweets is informative and reflects the real-time events. Interestingly, in the word clouds visualization, we can notice that the conversations surrounding COVID-19 in all three countries have a political aspect to them. Words such as state and government, political leaders such as Trump, Boris Johnson and Modi are all frequently present. Future research should explore the correlation of these terms and their sentiment scores across COVID-19 events.

Applying unsupervised learning method and topic modeling in particular shows quite informative and accurate results. The two research questions raised were answered through a detailed analysis in which we obtain an understanding of the public topics discussed during the COVID-19 pandemic using the three months of related tweets in the UK, India and the USA. In addition to the topics, the daily mean public sentiment of tweets related to the government topic gives us an understanding of public opinions regarding their government actions.

This study is not without limitations, however. Debate exists as to the efficacy of using Twitter data (Gayo-Avello *et al.*, 2011). Twitter users may not be representative of the general population, not even as a "sample" of the general populace. This issue is exacerbated in a developing country such as India where relatively few have access to the internet. Questions surround the nature of citizens' use of Twitter – how many users are original posters and re-tweeters? Should an original post carry more weight than its retweet by others? What about the general bias of tweets? Some argue that tweets or social media are more negative in general, whereas others argue that positive tweets are retweeted more often (Ferrara and Yang, 2015). All of these issues could potentially cloud the use of Twitter for this study and those of others. However, despite these limitations, Twitter is being accepted as a valid indicator of popular sentiment and information by the press and other institutions.

7. Conclusion

In this paper, we perform topic modeling and sentiment analysis of Twitter data during a three-month period. We discover that governments are a major COVID-19–related topic of discussion on Twitter, and moreover, users' sentiments about their governments are country specific.

We discover differences in public sentiment on twitter toward COVID-19 pandemic in the three countries included in our analysis. For instance, the most dominant topic of discussion in terms of number of tweets was Government in the UK. However, the sentiment of discussion was neutral overall for this topic, indicating that the people were not dissatisfied with British Government's response to the pandemic. On the other hand, government is not the most dominant topic of discussion on Twitter in the USA and India.

In the USA, the most frequently discussed topic related with the origins of COVID-19 and contains tweets discussing various conspiracy theories. It is well documented that social media played a huge role in propagating conspiracy theories during the COVID-19 pandemic (Ferrara, 2020). In fact, these conspiracy theories were so prevalent on social media that COVID-19 has been labeled as an "Infodemic" by some researchers (Cinelli *et al.*, 2020; Diseases, 2020). We also observe interestingly that in the case of the USA, the second most popular topic contains terms such as *Donald Trump*, *protest*, *riot*, *loot* and *crises* indicating the divisiveness and political nature of COVID-19 in the country. President Donald Trump was under the microscope in terms of his response to the pandemic where he initially downplayed the severity of the virus and later had to change his tone and response. We also observe from the keywords in this topic that there were various protests during this time period as many citizens did not agree with the government restrictions placed during the pandemic. Interestingly, research studies have indicated political ideology as playing a very important role in susceptibility to fake news (Calvillo *et al.*, 2020). It is also interesting to note here that the USA had the most negative sentiment among all countries when it comes to Twitter discussion related with COVID-19.

The political aspect of the pandemic was visible in all three countries as words mentioning government along with Trump, Johnson and Modi, heads of states of the USA, the UK and India, respectively, occurred frequently in the discussion. This indicated that populace was observing the leadership in determining the course of action during this crisis. This phenomenon has been observed by various other studies looking at the COVID-19 pandemic, highlighting the role of leadership in responding to the COVID-19 pandemic along with political and governance dimension of the crises (Dodds *et al.*, 2020; Al Saidi *et al.*, 2020).

This study demonstrates the usefulness and efficacy of combining knowledge extraction (topic modeling) with sentiment analysis to explore and analyze a social media source such as Twitter. Topic knowledge informs sentiment, whereas sentiment colors the nature of topic, together forming as a powerful model to analyze citizen opinion toward a pandemic. Governments and policymakers can better understand the sentiment of their citizens regarding certain actions. Analysts, think tanks and pundits can use this information to motivate additional research.

Notes

1. https://drive.google.com/drive/folders/15tVt6xR0pRmUIVPk5Vd62L1_5KKoNQJd
2. https://pypi.org/project/gensim/

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